cosmological constant—left unspecified. We now realize that quantum physics suffuses empty space with jittering fields, and as we directly see through Casimir's discovery, the resulting microscopic field frenzy fills space with energy. In fact, a major challenge facing theoretical physics is to show that the combined contribution of all field jitters yields a total energy in empty space—a total cosmological constant—that is within the observational limit currently determined by the supernova observations discussed in Chapter 10. So far, no one has been able to do this; carrying out the analysis exactly has proven to be beyond the capacity of current theoretical methods, and approximate calculations have gotten answers *wildly* larger than observations allow, strongly suggesting that the approximations are way off. Many view explaining the value of the cosmological constant (whether it is zero, as long thought, or small and nonzero as suggested by the inflation and the supernova data) as one of the most important open problems in theoretical physics.

6. In this section, I describe one way of seeing the conflict between general relativity and quantum mechanics. But I should note, in keeping with our theme of seeking the true nature of space and time, that other, somewhat less tangible but potentially important puzzles arise in attempting to merge general relativity and quantum mechanics. One that's particularly tantalizing arises when the straightforward application of the procedure for transforming classical nongravitational theories (like Maxwell's electrodynamics) into a quantum theory is extended to classical general relativity (as shown by Bryce DeWitt in what is now called the Wheeler-DeWitt equation). In the central equation that emerges, it turns out that the time variable does not appear. So, rather than having an explicit mathematical embodiment of time—as is the case in every other fundamental theory—in this approach to quantizing gravity, temporal evolution must be kept track of by a physical feature of the universe (such as its density) that we expect to change in a regular manner. As yet, no one knows if this procedure for quantizing gravity is appropriate (although much progress in an offshoot of this formalism, called *loop quantum gravity*, has been recently achieved; see Chapter 16), so it is not clear whether the absence of an explicit time variable is hinting at something deep (time as an emergent concept?) or not. In this chapter we focus on a different approach for merging general relativity and quantum mechanics, *superstring theory.*